



Credit Card Fraud Detection Analysis

A data-driven approach to minimizing financial loss while maintaining customer trust

AUTHOR: MICHELLE MWENDE



Business Challenge

Financial Risk

Credit card fraud causes direct monetary losses and erodes customer trust

Competing Priorities

Balance detecting fraud (false negatives) vs. blocking legitimate transactions (false positives)

Legacy Systems

Static rules struggle to adapt to evolving fraud patterns

Project Objectives

01

Explore Transaction Patterns

Understand fraudulent vs. non-fraudulent behavior through EDA

03

Build Classification Models

Compare multiple approaches for identifying fraud

02

Address Class Imbalance

Assess impact on fraud detection performance

04

Optimize Performance

Balance false positives and false negatives to minimize expected loss



Dataset Overview

284,807 Transactions

European cardholders over two days in September 2013

- 31 numerical features (V1-V28 PCA-transformed, Time, Amount)
- Highly anonymized for confidentiality
- Clean dataset with no missing values

0.17%

Fraudulent

Only 492 fraud cases

578:1

Imbalance Ratio

Extreme class imbalance

The Imbalance Challenge

With 99.83% legitimate transactions, traditional accuracy metrics are misleading. A model predicting all transactions as legitimate would achieve 99.83% accuracy while detecting zero fraud.





Data Preparation Strategy

1

Feature Scaling

Standardized Time and Amount features while preserving PCA-transformed variables

2

Train-Test Split

80-20 split with stratification to maintain fraud distribution

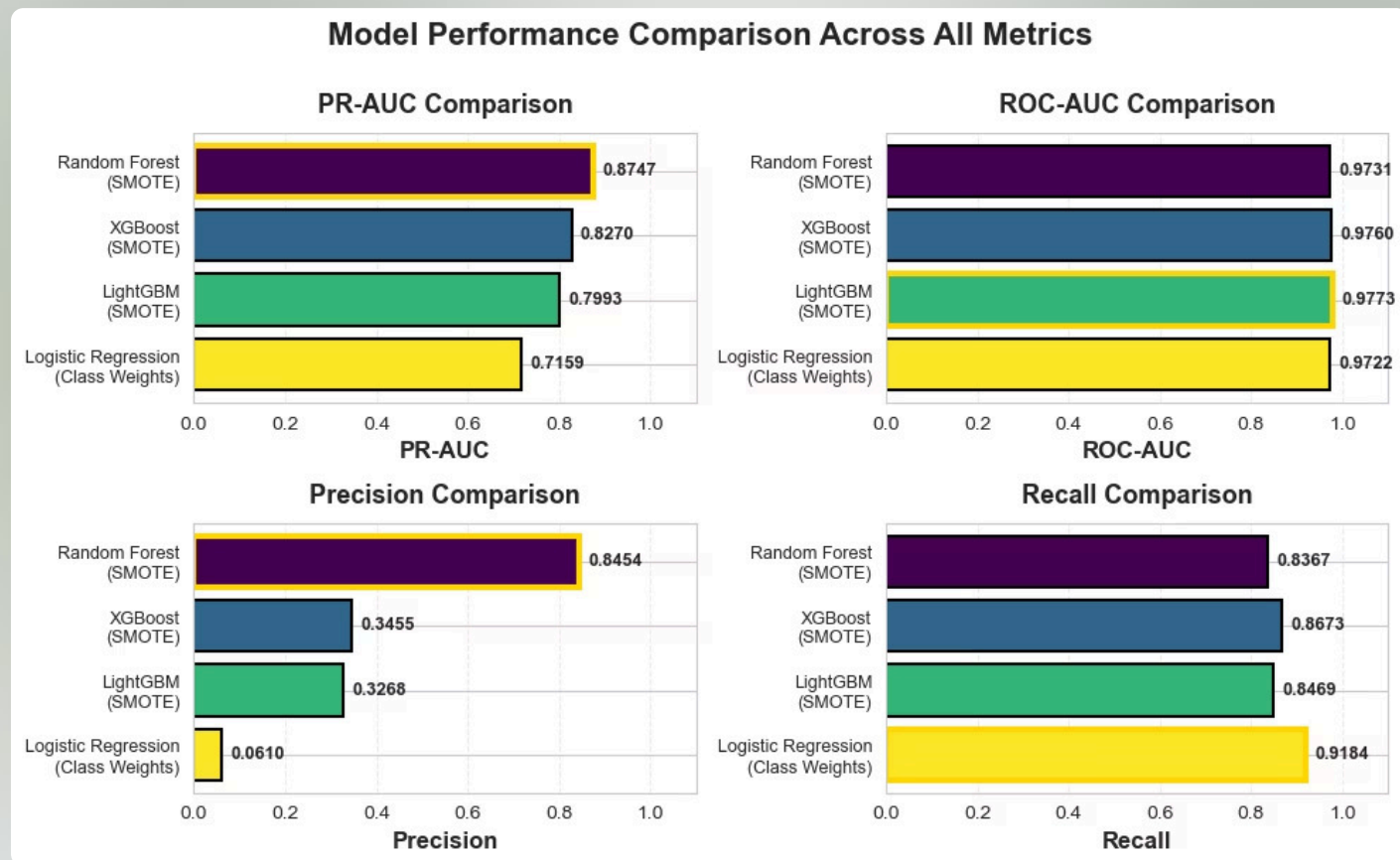
3

Imbalance Handling

Tested SMOTE, undersampling, and class weights

Model Performance Comparison

Random Forest with SMOTE emerged as the clear winner, achieving the highest PR-AUC of 0.8747 and best balance between precision and recall.



The Winning Model

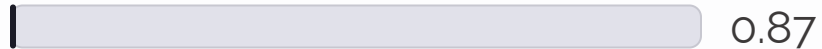
Random Forest + SMOTE



Precision



Recall



PR-AUC

Why This Model Wins

- Only 15 false positives vs. 1,386 for Logistic Regression
- Caught 82 out of 98 fraudulent transactions
- Optimal balance minimizes customer disruption
- Highest PR-AUC indicates superior performance on imbalanced data

Key Findings & Business Impact

Feature Importance

V14, V10, V4, V12, and V17 are the primary fraud indicators, accounting for over 60% of predictive power

Threshold Optimization

Lowering threshold to 0.35 increased recall to 89.8% while maintaining 76.5% precision—catching 11% more fraud

Operational Efficiency

Reduced false alarms by 99% compared to baseline, dramatically lowering manual review costs

Customer Experience

Minimized legitimate transaction blocks while maintaining robust fraud protection



Recommendations



Deploy Random Forest Model

Implement with 0.35 threshold for optimal fraud detection and customer experience balance



Continuous Monitoring

Track model performance and retrain quarterly as fraud patterns evolve



System Integration

Integrate with real-time transaction processing for immediate fraud flagging



Cost-Benefit Analysis

Regularly assess financial impact and adjust threshold based on business priorities

Thank You

Your time and attention are greatly appreciated. We hope this analysis provides valuable insights into enhancing our fraud detection capabilities and protecting our customers. For any questions or further discussion, please feel free to reach out.

Michelle Mwende

